

Anemia Detection from Fingernail RGB Images Using CNN

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Abstract—Anemia is a global health problem and is generally diagnosed through invasive laboratory tests that require special costs and facilities. This study proposes a nail image-based anemia detection using transfer learning with three CNN architectures, including DenseNet169, InceptionV3, and Xception, in a non-invasive manner. The dataset used is a combination of two publicly available sources with 4,466 nail images of anemia and non-anemia classes based on hemoglobin levels. Image pre-processing includes exposure normalization with CLAHE, nail area extraction via bounding box, YCbCr segmentation, and online data augmentation. The evaluation results on 670 test data showed that InceptionV3 achieved the best performance with an accuracy of 91.94% and AUC-ROC of 97.03%, followed by DenseNet169 with an accuracy of 90.90% and AUC-ROC of 95.50%, and Xception with an accuracy of 80.45% and AUC-ROC of 87.47%. These findings show the promise of a nail-image-based approach using transfer learning as a cost-effective and accessible approach to anaemia screening, especially in areas with limited health facilities.

Keywords—Anemia Detection, Fingernail Image, Convolutional Neural Network, Transfer Learning, DenseNet169, InceptionV3, Xception

I. INTRODUCTION

Anemia is a major health problem and is characterized by hemoglobin (Hb) levels below 11g/dL, which inhibit the distribution of oxygen and impact vulnerable groups such as pregnant women, toddlers, and informal workers. The World Health Organization (WHO) declares anemia a serious health problem in developing countries that contributes to decreased productivity [1]. In Indonesia, anemia detection is generally performed by the Complete Blood Count (CBC) test, which is invasive, requires laboratory facilities, trained personnel, and costs Rp 100,000–300,000 per examination.

In rural areas such as Depok, limited access to laboratories, lack of testing supplies, and transportation problems cause a misdiagnosis rate of up to 50%, resulting in national economic losses of about Rp 50 trillion annually. Paleness at the nail

base is a relatively reliable noninvasive clinical sign for severe anemia with a sensitivity of 70 – 80%, but manual visual inspection only achieves 50 – 70% accuracy due to observer bias, exposure variations, and differences in skin tone.

Recently, there has been work on RGB-based nail image analysis for the detection of noninvasive anemia. The public datasets, such as the Ghana/Mendeley dataset, offer 92 – 99% accuracy under controlled conditions using nail image segmentation, color space conversion (LAB, YCbCr), and CNN [2]. The Naïve Bayes Classification with YCbCr segmentation was reported with 87.3% accuracy in a 2025 study published in Frontiers Media [3]. However, there are several challenges, including sensitivity to exposure differences, overfitting, low recall in borderline Hb levels (10–12g/dL), and interpretability of the models.

As a solution, a CNN-based deep learning approach was developed with standard preprocesses such as resizing, lighting normalization, and extraction of the base region of the nail. Three transfer learning architectures are used: DenseNet169 to capture subtle RGB color variations, InceptionV3 for multiscale feature extraction, and Xception for compute efficiency through depthwise separable convolution. All three models were refined on the nail dataset and evaluated using accuracy, precision, recall, F1-score, and AUC-ROC with cross-validation.

II. LITERATURE REVIEW

The conjunctiva is the most studied area of the body for non-invasive anemia screening, among other areas of the body, due to its visibility and ease of imaging with smartphones. The ML approach based on ensemble stacking with CIELAB's color features achieved an accuracy of 89.4% and AUC of 0.97 [4], [5]. The color ratio and regression-based methods reported the moderate accuracy of 75.4%–82.9% [6], [7], [8]. Deep learning development achieved an accuracy of 98.47%

and 97.94% for ViT-B16 and ResNet-50, respectively [9],[10]. In the mobile implementation, YOLOv8+KNN achieved 83% accuracy on the Android platform [11] while a multimodal approach combining conjunctival imagery and EHR data achieved 95% [12]. However, variations in lighting conditions and a lack of standardized image acquisition protocols remain major challenges.

Fingernails are also widely studied because of their ease of imaging and potential as a self-screening method. CNN architectures such as Fast R-CNN, EfficientNetB0, and DenseNet169, as well as decision trees based on RGB color features, consistently achieve an accuracy of 89%–94% [13], [14], [15], [16]. Meanwhile, the RGB color feature-based decision tree reported a similar accuracy of around 90% for pre-diagnosis needs [17]. Large-scale implementation by Mannino et al. on 1.4 million application uses with a sensitivity of 89% proves the feasibility of this approach in real-world scenarios [18]. However, the skin color bias, lighting variability, and dataset size limitations are still important challenges [2], [19]. [20] also contributed a benchmark dataset to facilitate objective comparisons across methods.

Beyond the fingernail, the palm also garnered significant research attention for its clear and consistent color contrast and ease of imagery. Several CNN-based, decision tree, and random forest methods have shown accuracy above 99%, particularly in child populations in resource-constrained environments [21], [22], [23], with a multimodal approach combining palm imagery and clinical data achieving 99.58% [24]. More research is needed, however, to extend to adult populations and demographic diversity.

Some studies also explore other areas of the body as well as multi-research approaches. High accuracy was reported on a lip-based approach using Xception (99.28%) [25], MobileNet-based multi-area (99%) [26], palm-based with Naive Bayes (99.96%) [27], as well as a systematic review showing the ensemble method was capable of reaching 99%–100% [28]. Meanwhile, face-based detection showed a more moderate accuracy of 84% [29]. Despite these results, the lack of standardized datasets is still a major obstacle in the evaluation between methods [30], [31]. [32] explored a blood cell image-based approach as an alternative beyond external body areas, also yielding promising results.

III. METHODOLOGY

In this study, a deep learning approach is used to detect anemia non-invasively from nail images. The research flow includes data collection, dataset preparation, image processing, data splitting, CNN model training, and evaluation. The main objective of this research is to develop a robust model that is able to detect anemia from the nail image despite the variation in lighting and image quality. The entire process of the present study is described in Fig. 1.

A. Data Collection and Preparation

The two-nail public dataset used in this study. The first dataset is obtained from the “Detection of Anemia using

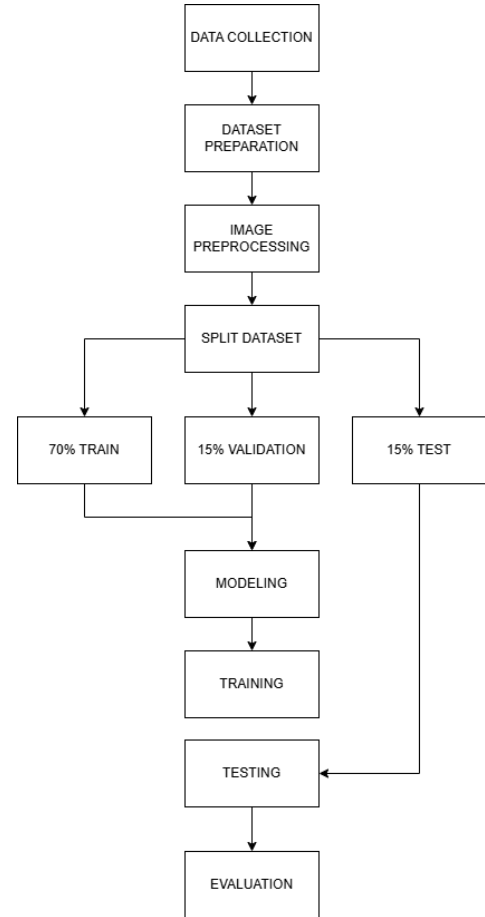


Fig. 1. Research Workflow.

Colour of the Fingernails Image Datasets from Ghana” [15]. The dataset contains 710 original images (426 anemic + 284 non-anemic) and was expanded to 4,216 images with augmentation. The images were collected in controlled conditions and the label of anemia was determined by the hemoglobin results from the laboratory.

The second dataset is the “Dataset of Human Skin and Fingernails Images for Non-Invasive Hemoglobin Level Assessment” by [20] that contains RGB images of fingers from 250 patients with known Hb levels (HB_LEVEL_GperL) with values recorded in units of g/L automatically converted to g/dL, balanced by sex (128 males, 122 females), spanning the age range of 18–95 years. Anemia labels were assigned based on Hb values (Hb threshold <11.0 g/dL), resulting in 43 anemic samples and 207 non-anemic samples.

After the dataset is obtained, the next step is data preparation. The images were divided into two class label datasets: anemia and non-anemia. Augmentation is applied online (on-the-fly) during the training process using a data generator, including rotation, horizontal flip, brightness adjustment, and noise addition are applied to improve variation and reduce overfitting. Because augmentation is performed dynamically during training, the number of files on the disk remains the

same as the number of original images. The distribution of the combined dataset is presented in Table 1.

TABLE I
DATASET DISTRIBUTION

Category	Images Count
Anemic ($Hb < 11$ g/dL)	2564
Non-Anemic ($Hb \geq 11$ g/dL)	1902
Total	4466

B. Image Preprocessing and Data Splitting

Before training, images are processed to make the input standardize. Lighting normalization is performed to reduce lighting variations using CLAHE (Contrast Limited Adaptive Histogram Equalization) in LAB color space. Two approaches are employed for nail area extraction: using bounding box coordinates from CSV metadata if available, or YCbCr color space-based segmentation if not. All images are resized to 224×224 pixels according to standard CNN input, and pixel values are normalized to the range $[0, 1]$ by dividing by 255.

Afterwards, the data set is divided into three sets: training data, validation data, and testing data. 70% of the data is used for training to optimize the model parameters, 15% is used for validation to tune hyperparameters, and perform premature termination. The remaining 15% is used as test data to assess the model's final performance on data that has never been seen before. With a total of 4,466 images, this division results in approximately 3,126 training data, 670 validation data, and 670 test data.

C. Model Training, Testing and Evaluation

After the preprocessing and dataset splitting phases are complete, the model is trained using the training data. Transfer learning is implemented using three pre-trained CNN architectures: DenseNet169, InceptionV3, and Xception. This architecture was selected due to its complementary strength in feature extraction as shown in TABLE 2. Those models were trained using adam optimizer with learning rate of 1×10^{-4} , a categorical cross-entropy loss function, a batch size of 16, a maximum of 50 epochs, and early stopping to prevent overfitting were used for training the models. The training process consisted of two phases: the feature extraction phase (base model layers frozen) followed by the fine-tuning phase (some of the top layers unfrozen for retraining).

TABLE II
CNN ARCHITECTURE COMPARISON

Architecture	Strengths
DenseNet169	Strong feature reuse and sensitivity to color variations
InceptionV3	Effective in capturing multi-scale visual features
Xception	Efficient with lower computational complexity

Following the training phase, the final testing stage was conducted using test data (15% of the total dataset, about 670

images) which is completely separate from the training and validation data. This step is used to evaluate how well the model can generalize to new data that has not been seen during the training or validation process.

To assess model performance, several evaluation were applied with different classification metrics: accuracy, precision, recall, F1-score, and AUC-ROC as summarized in Table 3. Furthermore, a confusion matrix is used to analyze the classification outcome in more detail.

IV. RESULT AND DISCUSSION

TABLE III
COMPARISON OF MODEL PERFORMANCE

Metric	DenseNet169	InceptionV3	Xception
Accuracy	90.90%	91.94%	80.45%
Precision	90.90%	91.94%	80.48%
Recall	90.90%	91.94%	80.45%
F1-Score	90.90%	91.92%	80.23%
AUC-ROC	95.50%	97.03%	87.47%
Anemic Precision	92.19%	92.11%	80.24%
Non-Anemic Recall	89.47%	89.12%	70.88%
Training Stability	Good	Best	Moderate

We used the same dataset to test DenseNet169, InceptionV3, and Xception (385 anemias, 285 non-anemias), with identical preprocessing and training. As shown in TABLE 4, inceptionV3 performs the best with an accuracy of 91,94%, precision, recall, and F1-score above 91%, which show a consistent and balance classification in both classes. Also, inceptionV3 achieved the highest anemia recall 94,03% with only 23 false negatives. This is critical in medical screening because missing true anemia cases has more serious clinical implications than false positives. DenseNet169 achieved an accuracy of 90,90% with the AUC-ROC of 95,50%, which show good discriminating capabilities at difference classification thresholds. Meanwhile, Xception had the lowest performance with an accuracy of 80,45% and an AUC-ROC of 87,47%. Both DenseNet169 and InceptionV3 demonstrate stable and consistent convergence behavior during the feature extraction and fine-tuning stages. On the other hand, Xception has a very sharp spike in loss when transitioning to the fine-tuning phase around the 20th epoch, which is likely to affect its overall performance on the test set.

Mostly, the results of this study generally have similarities with previous studies, although some differences in performance can still be found. DenseNet169 in this study was higher than [2] which only reached 74.09% for AUC-ROC also accuracy, recall, and overall F1-score below 72%, on a smaller dataset (909 images), suggesting that the merging of both datasets into 4,466 images and the application of CLAHE also online augmentation contributed to better model generalization. Also, [16] achieved 89% accuracy using DenseNet169 on a similar fingernail dataset, which is slightly lower than the 90.90% in this study.

On the other hand, [13] Fast R-CNN with an accuracy of 92.8% with a region-based detection approach on 2,240

augmented images outperformed all three models in the study, showing that object detection methods focusing on specific areas of interest might be more beneficial to capturing discriminative basic features of the nails. The approach in [14] achieves 94% accuracy using EfficientNetB0 with RGB and texture-based features, showing that architecture choice plays a big role in performance for this classification task.

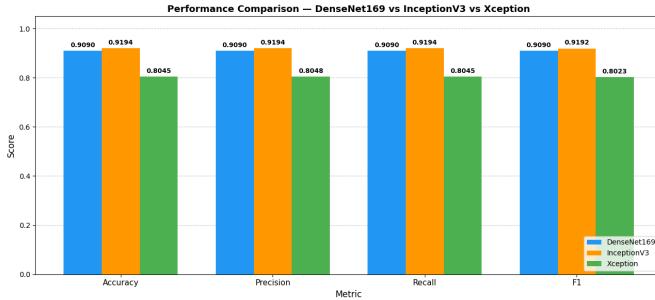


Fig. 2. Performance Comparison — DenseNet169 vs InceptionV3 vs Xception.

Overall, this comparison show that InceptionV3 with an accuracy of 91,94% and AUC-ROC of 97,03% ,as shown in Figure 2, provides a robust foundation for nail-based anemia detection. Moreover, the AUC-ROC score of 95,50% achieved by DenseNet169 in this study is also worth mentioning, as the ability to discriminate at various threshold is often more important in clinical screening applications than the raw accuracy value.

There are some limitations in this study. First, even though the dataset has been increased to 4,466 images, the class distribution is still unbalanced (2,564 anemia vs. 1,902 non-anemia), which reduces the model performance on the non-anemia class as can be seen by the non-anemia recall of Xception only achieved 70,88%. Moreover, the second dataset only has 250 images with imbalanced class ration, so its contributions to diversity of anemic class representation is still limited. The dataset problem may not fully represent real-world variations such as skin tone, lighting conditions, and nail appearance. The binary classification used may also face difficulties where hemoglobin levels close to the 11 g/dL threshold and the visual difference between classes are very subtle. The lack of explainability methods, such as Grad-CAM, makes it difficult to verify if the model is focusing on clinically relevant nail.

V. CONCLUSION

This study successfully developed a non-invasive anemia detection system using nail imagery with three transfer learning architectures: DenseNet169, InceptionV3, Xception. These architectures trained on a combined dataset of 4,466 images from two public sources. From the evaluation results, it was found that InceptionV3 performed best overall, as shown in Figure 2, with an accuracy of 91.94% and an AUC-ROC of 97,03% , followed by DenseNet169 with an accuracy of 90.90% and an AUC-ROC of 95,50% , whereas Xception

performed with an accuracy of 80.45% and an AUC-ROC of 87,47% . The combination of two different public datasets along with CLAHE-based lighting normalization and online data augmentation was shown to contribute to improving model generalization as compared to previous studies with a single dataset of smaller size.

For further research, it is recommended to explore additional architectures, such as EfficientNet and Vision Transformers, to further improve performance. In addition, the application of explainability methods, such as Grad-CAM, can help to confirm the clinical relevance of the features learned by the model. Further expansion of datasets with more diverse skin colour variations and lighting conditions and model validation in real clinical settings represent an important step before this system can be practically applied as an anaemia screening tool.

AUTHOR CONTRIBUTIONS

H. Humayra, V. Novalisa, and E. Simin carried out the research, including formulating the study design, implementing the transfer learning models for nail image-based anemia detection, running the experiments, and analyzing the results. M. Nasari and S. Ahmad took charge of constructing and confirming the study's methodology, supervising its overall progress, and managing administrative aspects of the project. Manuscript drafting and revision were carried out collaboratively by all authors.

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